

# ThoughtFlow: A Negative-Result Falsification Ladder for Sparse-Cache Retention

**Anonymous authors**

Paper under double-blind review

## Abstract

We present a preregistered falsification protocol for training-free sparse KV-retention proposals on Mac-local saved sparse-cache fixtures. The contribution is diagnostic, not a positive method: freeze a sparse-cache quality surface, preregister each successor signal before measurement, use matched-budget proxy rows as promotion tests and stop signals that fail them, separate stopped-family rows from proxy-baseline rows, and report paired uncertainty plus oracle/headroom diagnostics where those diagnostics are implemented. All benchmark surfaces here are Mac-local saved-trace falsification fixtures, not reasoning-model benchmark claims. As a case study, the original anchor/recent/phase/math family is stopped, recurrence-distance utility (`rdu_topk`) clears one frozen 74-trace surface but fails stricter reproduction, prefix-surprisal utility (`psi_topk`) fails on a fresh 70-trace C2C GSM surface, and value-weighted attention contribution (`vwac_topk`) fails on a fresh 64-trace C2C SVAMP surface. The original FP8 systems target is not part of this claim: the anchor/phase int8 quantization primitive passes Triton interpreter correctness tests against its CPU reference, but no real FP8, CUDA, latency, throughput, or live compression-method claim is made.

## 1 Contribution

The contribution is a reusable stop-rule protocol, not another sparse-cache policy. The protocol combines seven checks: no-retune frozen surfaces, pre-registered successor utilities, matched-budget proxy baselines, stopped-family/proxy-baseline separation, paired uncertainty, oracle/headroom diagnostics for the RDU demotion gates, and a tracked diagnostic packet that hashes the stop artifacts. ThoughtFlow-FP8 is historical naming only: this draft evaluates sparse-cache falsification, not an FP8 serving method. It is the case study showing why each check matters: one plausible signal appears positive on an initial frozen surface, then fails to reproduce under stricter reproduction, while two later signals fail fresh-surface gates outright. If older tracked artifacts say ALIVE, REPRODUCED, or PROMOTED, the current decision manifest supersedes them.

Reviewer takeaway: the protocol is designed to stop attractive false positives before they become GPU or kernel projects. In this case it catches a first-surface `rdu_topk` win, forces same-family and proxy-baseline separation, and then uses independent/fresh saved-trace surfaces to demote or kill every consumed successor without retuning.

The next table is included only to show why the branch looked promising before the fresh gates. It is not a current leaderboard: every candidate policy row in the table is superseded by the demotion ladder that follows.

## 2 Signals Under Test

The original policy attempted to protect anchors and phase labels. The first successor adds a small recent-token reserve. The second successor attempted to protect anchors, reserve half the retained budget for recent tokens, and let phase markers compete with math-state

| Protocol check      | Purpose                                                     | Stop condition                                                                                                                                                               |
|---------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frozen surface      | Prevent post-hoc tuning.                                    | A successor is measured once after preregistration.                                                                                                                          |
| Matched budget      | Avoid full-cache comparisons.                               | Must beat R-KV-like and ThinKV-like rows at the same keep rate.                                                                                                              |
| Stopped/proxy split | Separate local family reuse from proxy-baseline robustness. | A stopped-family-only win is diagnostic, not a method claim.                                                                                                                 |
| Paired intervals    | Avoid mean-only promotion.                                  | For $\Delta = \text{NLL}_{\text{candidate}} - \text{NLL}_{\text{baseline}}$ , promotion requires mean $\Delta \leq -0.03$ and paired CI high below zero versus both proxies. |
| Oracle/headroom     | Show whether a useful policy exists at the budget.          | Large oracle gaps block method claims.                                                                                                                                       |
| Fresh surface       | Test robustness.                                            | Failure demotes or kills the signal.                                                                                                                                         |

Table 1: Falsification protocol used for each sparse-cache successor.

| Signal        | Final surface                                                                     | Final state                                            |
|---------------|-----------------------------------------------------------------------------------|--------------------------------------------------------|
| anchor family | 74-trace frozen sparse-cache gate with matched ThinKV-like and R-KV-like proxies. | Stopped: no robust matched-budget win.                 |
| rdu_topk      | Alternate surface plus independent 89-trace saved-trace reproduction.             | Killed: first-surface pass fails reproduction.         |
| psi_topk      | Fresh 70-trace C2C GSM saved-trace surface.                                       | Killed: large NLL regression versus both proxies.      |
| vwac_topk     | Fresh 64-trace C2C SVAMP saved-trace surface.                                     | Killed: worse than both proxies with paired intervals. |

Table 2: Final status matrix. The contribution is the auditable stop protocol and its failure modes, not a live sparse-cache method.

and high-importance reasoning tokens under a saliency bonus. Those policies are now stopped on the current saved traces. The pre-registered `rdu_topk` successor is demoted after stricter reproduction; later `psi_topk` and `vwac_topk` successors are killed by fresh-surface gates.

The proxy rows in Table 3 are deliberately simple matched-budget controls, not faithful reimplementations of the cited systems: R-KV-like keeps the first four anchor tokens plus the most recent tail, ThinKV-like keeps a recent reserve plus a math-state importance bonus, and LongFlow-like is an importance top- $k$  retention rule. The “-like” suffix is part of the local row name: each such row is a Mac-local proxy, not a benchmark comparison to the named method. The local definitions live in `experimental/thoughtflow_fp8/phase2/simulate_phase_retention.py`; the paper uses them only as falsification controls for the stopped signal family.

### 3 Hidden/KV Telemetry

We then ran a Mac-local telemetry gate using real `distilgpt2` tensors rather than only text markers. The probe ranks retained tokens by attention received, final-hidden norm, key norm, value norm, or combined KV norm. This is still not sparse-KV decoding, but it checks whether the interpretable phase policy is merely preserving labels that hidden/KV saliency would not select.

Paired against `value_norm_topk`, ThoughtFlow’s phase-recall delta is +0.508 with 95% CI [+0.436, +0.579]. The math-state delta is only +0.073 with 95% CI [-0.078, +0.223]. This rules out a method claim based on phase recall alone: it may be marker preservation rather than useful cache retention.

| Policy                      | Keep rate | NLL   | $\Delta$ NLL vs full | PPL  |
|-----------------------------|-----------|-------|----------------------|------|
| full context                | 1.000     | 2.101 | 0.000                | 9.7  |
| R-KV-like                   | 0.210     | 3.419 | 1.319                | 38.7 |
| ThinKV-like                 | 0.210     | 3.583 | 1.482                | 44.6 |
| LongFlow-like               | 0.210     | 3.961 | 1.861                | 74.2 |
| ThoughtFlow                 | 0.210     | 3.961 | 1.861                | 74.2 |
| ThoughtFlow-recent          | 0.210     | 3.562 | 1.461                | 44.2 |
| ThoughtFlow-saliency-recent | 0.210     | 3.434 | 1.333                | 38.1 |
| Held-out R-KV-like          | 0.211     | 3.482 | –                    | 43.7 |
| Held-out sweep best         | 0.211     | 3.480 | –                    | 43.7 |

Table 3: Case-study retained-context continuation proxy on distilgpt2 traces. Lower NLL is better. R-KV-like, ThinKV-like, and LongFlow-like rows are Mac-local proxy baselines in this harness, not faithful implementations. Full context is not a matched-budget baseline. The held-out sweep row is selected on 12 train traces and reported on 12 held-out traces, so its NLL is not directly averaged with the all-trace rows.

| Policy               | Keep rate | Anchor | Phase | Math-state |
|----------------------|-----------|--------|-------|------------|
| ThoughtFlow          | 0.207     | 1.000  | 1.000 | 0.918      |
| LongFlow-like        | 0.207     | 1.000  | 1.000 | 0.918      |
| ThinKV-like          | 0.207     | 1.000  | 0.948 | 0.848      |
| value_norm_topk      | 0.207     | 0.000  | 0.492 | 0.845      |
| train-selected sweep | 0.207     | 1.000  | 0.430 | 0.956      |

Table 4: Hidden/KV saliency telemetry on 24 distilgpt2 traces. The strongest real-saliency proxy by phase+math recall is value\_norm\_topk. ThoughtFlow has higher phase recall than it, but LongFlow-like ties ThoughtFlow under the local importance heuristic.

## 4 Sparse-Cache Case Study Gate

We next ran a CPU-only sparse-cache probe. The model processes the full prefix once, each policy prunes the returned KV cache to the same 0.20 budget, and the continuation is scored from the pruned cache. This is not a Triton, CUDA, FP8, latency, or throughput result. It is closer to the cache-use failure mode than retained-text NLL, but still CPU-only diagnostic evidence.

| Policy                      | Keep rate | NLL   | Paired $\Delta$ vs R-KV-like |
|-----------------------------|-----------|-------|------------------------------|
| full cache                  | 1.000     | 2.142 | -1.296 [-1.533,-1.066]       |
| ThoughtFlow-saliency-recent | 0.210     | 3.372 | -0.067 [-0.151,+0.011]       |
| ThinKV-like                 | 0.210     | 3.389 | -0.049 [-0.192,+0.077]       |
| ThoughtFlow-recent          | 0.210     | 3.399 | -0.040 [-0.169,+0.074]       |
| ThoughtFlow sweep best      | 0.210     | 3.436 | -0.002 [-0.007,+0.000]       |
| R-KV-like                   | 0.210     | 3.438 | 0.000                        |
| LongFlow-like               | 0.210     | 3.588 | +0.150 [-0.016,+0.300]       |
| ThoughtFlow                 | 0.210     | 3.588 | +0.150 [-0.006,+0.297]       |

Table 5: CPU sparse-KV continuation quality on 24 distilgpt2 traces. R-KV-like, ThinKV-like, and LongFlow-like rows are Mac-local proxy baselines, not faithful implementations. Negative paired delta is better than R-KV-like. ThoughtFlow-saliency-recent is best in mean NLL, but it clears the strongest non-ThoughtFlow row by only 0.017 NLL and remains below the pre-registered 0.03 promotion margin.

A bounded sparse-cache sweep over 24 nearby anchor/recent/phase/math configurations selects tf\_sparse\_r0.55\_p0.05\_m0.12\_a2 on the 12 train traces. On the 12 held-out traces it gets NLL 3.340 versus ThinKV-like 3.385 and R-KV-like 3.420. This clears the mean 0.03 margin, and the paired delta versus R-KV-like is -0.080 with 95% CI [-0.152,-0.014]. However, the paired delta versus ThinKV-like is -0.045 with 95% CI [-0.226,+0.182]. The

fixed ThoughtFlow-saliency-recent incumbent has lower held-out mean NLL, 3.304, but its paired CIs also cross zero. This is a promising falsification candidate, not a method result.

We then froze ThoughtFlow-saliency-recent and `tf_sparse_r0.55_p0.05_m0.12_a2` and ran a larger 74-trace sparse cache slice with no retuning. ThinKV-like is the best compressed row among the stopped family comparison at NLL 3.900. The frozen sparse ThoughtFlow row gets 3.908, with paired delta -0.031 versus R-KV-like and 95% CI [-0.078,+0.020], but +0.008 versus ThinKV-like with 95% CI [-0.060,+0.085]. ThoughtFlow-saliency-recent gets 3.920. The larger no-retuning gate therefore stops this policy family.

After that stop decision, we pre-registered `rdu_topk`. It computes recurrence-distance utility from full-prefix prefill attentions using lag buckets [8,15], [16,31], [32,63], and [64,∞], then keeps the top budget by utility with ties broken by lower position. It does not use continuation tokens, trace labels, recent reserves, anchors, phase markers, or math-state bonuses. After `rdu_topk` failed reproduction, we pre-registered `psi_topk` (high self-surprisal prefix tokens) and `vwac_topk` (future attention weighted by cached value-vector norm).

## 5 Evidence Ladder

| Gate                                               | Readout                                                                                                                      | Decision           |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Synthetic retention                                | Anchor/phase labels can be protected at matched keep rate.                                                                   | Intuition only.    |
| Retained-text NLL                                  | Original ThoughtFlow rows do not beat the strongest proxy.                                                                   | Weakened.          |
| Hidden/KV telemetry                                | Phase recall improves, but math-state impact is not stable and LongFlow-like ties the heuristic.                             | Diagnostic.        |
| CPU sparse-cache probe                             | A tuned sparse ThoughtFlow row is promising in mean but not robust against ThinKV-like uncertainty.                          | Not promoted.      |
| Frozen 74-trace gate                               | <code>rdu_topk</code> beats ThinKV-like and R-KV-like on the first frozen surface.                                           | Candidate only.    |
| Same-slice rerun                                   | Cached result reproduces exactly on the deterministic Mac stack.                                                             | Repro bookkeeping. |
| Alternate surface                                  | Cross-family rows still lose, but a stopped same-family row beats <code>rdu_topk</code> by 0.006 NLL.                        | Weakened.          |
| Independent saved traces                           | R-KV-like is best compressed at NLL 3.981; <code>rdu_topk</code> reaches 4.014.                                              | Killed.            |
| Prefix-surprisal fresh saved-trace surface         | On distilgpt2 C2C GSM saved traces, <code>psi_topk</code> reaches NLL 7.899 versus ThinKV-like 3.906 and R-KV-like 3.960.    | Killed.            |
| Value-weighted attention fresh saved-trace surface | On distilgpt2 C2C SVAMP saved traces, <code>vwac_topk</code> reaches NLL 4.336 versus R-KV-like 4.096 and ThinKV-like 4.162. | Killed.            |

Table 6: ThoughtFlow-FP8 evidence ladder. The first-surface pass is shown as one step that failed to reproduce, not as the headline result.

The independent saved-trace check rules out claiming `rdu_topk` as a robust method on the available surfaces. A coarse trace-level decomposition explains where the reproduction fails: `rdu_topk` is +0.213 NLL worse than R-KV-like on long-prefix/high-RDU-density rows but -0.049 better on short-prefix/low-density rows. The large-regression buckets are exactly the buckets a robust policy would need to handle before any long-context claim.

**Prefix-surprisal successor.** A later consumed successor tested whether retaining high self-surprisal prefix tokens could preserve information that is hard to reconstruct. The signal uses only prefill logits, not continuation loss, trace labels, recent reserves, or RDU lag buckets. On a fresh 70-trace distilgpt2 C2C GSM saved-trace surface, `psi_topk` fails

| Registration | SHA-256 prefix | Measurement artifact                                                                                              |
|--------------|----------------|-------------------------------------------------------------------------------------------------------------------|
| rdu_topk     | 676d5239       | Same-surface pass, alternate-surface failure, and independent-trace failure are bundled in the diagnostic packet. |
| psi_topk     | cd8a6a10       | One-shot fresh distilgpt2 C2C GSM saved-trace sparse-cache check, artifact SHA prefix d0a08035.                   |
| vwac_topk    | a701495b       | One-shot fresh distilgpt2 C2C SVAMP saved-trace sparse-cache check, artifact SHA prefix 80a35954.                 |

Table 7: Registration-to-measurement ledger. Full hashes and artifact paths are recorded in the tracked diagnostic packet; this table exposes the audit chain inside the paper.

| Historical row     | Traces | Keep  | NLL   | $\Delta$ vs ThinkKV    | Later decision    |
|--------------------|--------|-------|-------|------------------------|-------------------|
| full cache         | 74     | 1.000 | 2.848 | -1.052 [-1.199,-0.919] | reference only    |
| rdu_topk           | 74     | 0.213 | 3.779 | -0.121 [-0.211,-0.037] | reproduction kill |
| ThinkKV-like       | 74     | 0.213 | 3.900 | 0.000                  | proxy baseline    |
| frozen sparse TF   | 74     | 0.213 | 3.908 | +0.008 [-0.060,+0.085] | stopped family    |
| TF-saliency-recent | 74     | 0.213 | 3.920 | +0.020 [-0.030,+0.074] | stopped family    |
| R-KV-like          | 74     | 0.214 | 3.939 | +0.039 [-0.015,+0.099] | proxy baseline    |
| LongFlow-like      | 74     | 0.213 | 4.158 | +0.258 [+0.178,+0.337] | proxy baseline    |

Table 8: Historical failed-to-reproduce one-shot frozen sparse-cache successor evaluation. R-KV-like, ThinkKV-like, and LongFlow-like rows are Mac-local proxy baselines, not faithful implementations. rdu\_topk also beats R-KV-like by -0.160 NLL with 95% CI [-0.264,-0.050] on this first surface, but later alternate-surface and independent-trace gates fail to reproduce the promotion.

decisively: ThinkKV-like is best compressed at NLL 3.906, R-KV-like reaches 3.960, and psi\_topk reaches 7.899. Paired deltas are +3.939 [+3.720,+4.144] versus R-KV-like and +3.994 [+3.773,+4.205] versus ThinkKV-like, where positive is worse for psi\_topk. This rules out this exact signal in the current Mac sparse-cache harness.

**Value-weighted attention successor.** The next consumed successor tested a cache-state utility: future prefix attention to a token multiplied by the token’s cached value-vector norm. On a fresh 64-trace distilgpt2 C2C SVAMP saved-trace surface, vwac\_topk also fails. R-KV-like is best compressed at NLL 4.096, ThinkKV-like reaches 4.162, and vwac\_topk reaches 4.336. Paired deltas are +0.241 [+0.030,+0.530] versus R-KV-like and +0.174 [+0.002,+0.392] versus ThinkKV-like.

## 6 Claim Boundary

**Limitations.** The ladder is demonstrated on Mac-local saved traces and CPU sparse-cache scoring, not native serving. The trace families are useful falsification surfaces, but they do not establish broad task coverage. The stopped signals also do not rule out future sparse-cache utilities; they only rule out the pre-registered policies under the reported surfaces and budgets. The scored surfaces use distilgpt2 saved continuations and saved prefix/KV telemetry as falsification fixtures; they are not reasoning-model benchmarks and not faithful external implementations of ThinkKV, R-KV, LongFlow, LazyEviction, ForesightKV, PM-KVQ, or KVQuant.

## 7 Stop Rule and Reopen Gate

The branch now contributes a diagnostic falsification ladder. The stopped anchor/recent/phase/math policy family remains ruled out as a tunable branch on this trace set, rdu\_topk fails reproduction, and psi\_topk/vvac\_topk fail their fresh-surface gates. Any

| rdu_topk check       | Readout                                                              | Decision            |
|----------------------|----------------------------------------------------------------------|---------------------|
| First frozen surface | 74 traces, NLL 3.779 versus ThinKV-like 3.900.                       | First-surface pass. |
| Same-slice rerun     | Exact deterministic reproduction; 0.145 NLL above compressed oracle. | Bookkeeping only.   |
| Alternate surface    | Same rule, NLL 3.594; a stopped same-family row reaches 3.588.       | Weakened.           |
| Independent traces   | 89 SVAMP traces; R-KV-like 3.981, rdu_topk 4.014.                    | Killed.             |
| Failure bucket       | +0.213 NLL on long-prefix/high-RDU-density rows.                     | Mechanism target.   |

Table 9: rdu\_topk demotion ladder. The first-surface pass is preserved as evidence that the protocol catches tempting positives, not as a live method claim.

| Lesson                                     | Observed failure                                                                    | Protocol implication                                        |
|--------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------|
| First-surface positives are weak evidence. | rdu_topk passed one frozen slice, then failed same-family and independent surfaces. | Require fresh-surface reproduction before promotion.        |
| Density regimes matter.                    | Long-prefix/high-RDU-density rows regress by +0.213 NLL.                            | Bucket diagnostics should be mandatory, not post-hoc color. |
| Oracle gaps prevent over-claiming.         | Same-slice reproduction remained 0.145 NLL above the compressed oracle.             | Report headroom before adding implementation work.          |
| Consumed signals should stay consumed.     | psi_topk and vwac_topk failed one-shot gates.                                       | Do not retune failed utility families on the same traces.   |

Table 10: Methodological lessons from the stopped sparse-cache ladder.

future positive-method claim requires a genuinely new pre-registered utility signal evaluated once on a fresh/larger frozen surface with strict stopped-family, proxy-baseline, and oracle/headroom reporting. The Triton interpreter gate is no longer the blocker for the quantization scaffold; method evidence is. This means no real FP8, CUDA, latency, or throughput result is claimed.

The positive value of the artifact is procedural: it reduces the chance that a plausible sparse-cache heuristic is promoted from a narrow frozen slice into implementation work before robustness exists. The stopped rows are therefore not failed leaderboard entries; they are worked examples of what the ladder is meant to prevent. The intended contribution is a reusable gate for future proposals, not a leaderboard entry for the stopped policies.

## 8 Related Work and Scope

This draft intentionally does not claim a new state-of-the-art KV-compression method. ThinKV, LongFlow, R-KV/R-KVHash-style retention, LazyEviction/ForesightKV-style future-use ideas, PM-KVQ, and KVQuant already occupy much of the sparse-cache and quantized-cache design space ThinKV (2026); LongFlow (2026); R-KV (2025); LazyEviction (2025); ForesightKV (2026); PM-KVQ (2025); KVQuant (2024). Our contribution is narrower: a Mac-local falsification ladder that shows how plausible interpretable retention signals can clear one frozen surface, fail stricter reproduction, or collapse on fresh surfaces. The project name retains “FP8” because that was the original systems target, but the current artifacts only validate CPU sparse-cache scoring and an int8/Triton-interpreter reference primitive; they do not measure FP8 numerical drift or native FP8 serving. The novelty is not preregistration or negative-results methodology in general; it is the domain-specific sparse-cache stop ladder with matched-budget proxy separation, stopped-family controls, and provenance locking.

| Claimed                                                                | Not claimed                                                            |
|------------------------------------------------------------------------|------------------------------------------------------------------------|
| Preregistered sparse-cache falsification ladder.                       | State-of-the-art KV compression.                                       |
| Matched-budget Mac sparse-cache scoring.                               | Native FP8 serving quality.                                            |
| Stopped-family/proxy-baseline and fresh-surface stop rules.            | CUDA kernel speed, latency, throughput, or HBM savings.                |
| Triton-interpreter parity for an int8 anchor/phase scaffold primitive. | A live deployable compression method or FP8/cache-serving correctness. |

Table 11: Boundary between the workshop claim and the stopped systems-method target.

## 9 Reproducibility

All reported gates use tracked scripts under `experimental/thoughtflow_fp8/phase2/` and the repo-local `./venv_arm64`. The frozen sparse-cache probe, same-slice rerun, alternate-surface rerun, independent saved-trace rerun, and fresh-surface successor checks each write Markdown and JSON artifacts with promotion criteria, paired intervals, and no-retuning decisions; oracle/headroom fields are present on the RDU demotion gates and are not claimed for the PSI/VWAC fresh successor packets. The tracked diagnostic packet hashes the saved JSON artifacts and records available model/source metadata plus input hashes for each saved artifact where those inputs are explicit. Legacy default trace paths are labeled as inferred from `run_real_trace_retention.DEFAULT_TRACES`. Its builder refuses regeneration when `experimental/thoughtflow_fp8/` is dirty, and the current manifest records a clean path status at generation. The hashed C2C fresh-surface inputs are tracked in the repository despite the root results ignore rule. The current package should still be read as a machine-checked falsification note rather than a full artifact-evaluation release. The stable local correctness command is:

```
./venv_arm64/bin/python \
  experimental/thoughtflow_fp8/phase2/build_diagnostic_packet.py
```

Then run:

```
TRITON_CPU_BACKEND=1 TRITON_INTERPRET=1 \
TRITON_HOME="$PWD/.debug/triton_home" \
./venv_arm64/bin/python -m pytest \
  experimental/thoughtflow_fp8/phase2/tests \
  experimental/thoughtflow_fp8/phase4/tests -q -rs
```

| Gate artifact    | Script                                                   | Input surface                         | Hash prefix |
|------------------|----------------------------------------------------------|---------------------------------------|-------------|
| frozen probe     | <code>frozen_sparse_cache_probe.py</code>                | frozen mixed traces, 74 scored        | f79c286c    |
| same-surface RDU | <code>rdu_no_retune_reproduction_check.py</code>         | same frozen traces, 74 scored         | 9a6420cf    |
| alternate RDU    | <code>rdu_alt_surface_reproduction_check.py</code>       | alternate measured surface, 73 scored | 1302ce31    |
| independent RDU  | <code>rdu_independent_trace_reproduction_check.py</code> | saved chat traces, 89 scored          | 5a568e1c    |
| psi_topk         | <code>psi_fresh_sparse_cache_check.py</code>             | distilgpt2 C2C GSM, 70 scored         | d0a08035    |
| vwac_topk        | <code>vwac_fresh_sparse_cache_check.py</code>            | distilgpt2 C2C SVAMP, 64 scored       | 80a35954    |

Table 12: Diagnostic-packet command map. Full commands, source metadata, and input SHA-256 values are in the tracked manifest. All rows are saved-trace proxy fixtures, not faithful LongFlow, ThinKV, or R-KV benchmark results.

## 10 Artifacts

Key artifacts are the current decision manifest, tracked diagnostic provenance packet, and reviewer pack: `experimental/thoughtflow_fp8/phase2/current_decision_manifest_20260506.md`, `experimental/thoughtflow_fp8/phase2/diagnostic_packets/thoughtflow_diagnostic_packet_20260506/manifest.json`, and `experimental/thoughtflow_fp8/paper/reviewer_pack.md`. The diagnostic packet records the RDU reproduction check, PSI/VWAC fresh-surface checks, README/progress handoff, workshop shell, and repo-local Triton interpreter note at `experimental/thoughtflow_fp8/phase4/triton_interpreter_note_20260506.md`.

## References

- ThinKV: Thought-Adaptive KV Cache Compression for Efficient Reasoning Models. ICLR, 2026. <https://openreview.net/forum?id=M3CeHnZKNC>.
- LongFlow: Efficient KV Cache Compression for Reasoning Models. arXiv:2603.11504, 2026. <https://arxiv.org/abs/2603.11504>.
- R-KV: Redundancy-aware KV Cache Compression for Training-Free Reasoning Models Acceleration. arXiv:2505.24133, 2025. <https://arxiv.org/abs/2505.24133>.
- LazyEviction: Lagged KV Eviction with Attention Pattern Observation. arXiv:2506.15969, 2025. <https://arxiv.org/abs/2506.15969>.
- ForesightKV: Optimizing KV Cache Eviction for Reasoning Models by Learning Long-Term Contribution. arXiv:2602.03203, 2026. <https://arxiv.org/abs/2602.03203>.
- PM-KVQ: Progressive Mixed-precision KV Cache Quantization for Long-CoT LLMs. arXiv:2505.18610, 2025. <https://arxiv.org/abs/2505.18610>.
- KVQuant: Towards 10 Million Context Length LLM Inference with KV Cache Quantization. arXiv:2401.18079, 2024. <https://arxiv.org/abs/2401.18079>.